

FedDiG: Frequency-Guided Diffusion Diversity for Generalizable Federated Time Series Classification

Haoran Shi*
Nanyang Technological University
Singapore, Singapore
SHIH0020@e.ntu.edu.sg

Junru Zhang*[†]
Zhejiang University
Hangzhou, China
junruzhang@zju.edu.cn

Cheng Peng
Zhejiang University
Hangzhou, China
chengchng@zju.edu.cn

Xiaoli Tang
Nanyang Technological University
Singapore, Singapore
xiaoli.tang@ntu.edu.sg

Longtao Huang
Alibaba Group
Hangzhou, China
kaiyang.hlt@alibaba-inc.com

Han Yu
Nanyang Technological University
Singapore, Singapore
han.yu@ntu.edu.sg

Abstract

Federated domain generalization (FDG) for time-series classification (TSC) poses a critical challenge for modern intelligent web services, which rely on edge-collected time-series signals from diverse mobile applications and web devices (e.g., wearables sensors) to support decision-making. The source heterogeneity and temporal dynamics give rise to out-of-distribution (OOD) patterns, which hinder the model's ability to generalize to previously unseen users and devices. In this work, we propose Federated Generalization via Diversity Generation (FedDiG), a diffusion-based FDG framework that captures intra-client distribution shifts from a frequency-domain perspective and employs cross-frequency sampling to synthesize time-series data with diverse spectral patterns. Specifically, FedDiG first performs frequency-proxy representation learning on clients to serve as diffusion conditions. The server then aggregates client-side frequency proxies to construct a global proxy pool and applies class-wise mixup to create novel frequency features. These features guide a global diffusion model to produce diverse data, enabling the simulation of previously unseen patterns and thereby enhancing model training. Extensive experiments on four cross-domain time-series benchmarks demonstrate that FedDiG significantly outperforms state-of-the-art federated learning and FDG baselines, particularly under small-data regimes and large-scale client scenarios, achieving robust generalization to unseen domains in federated settings. This work bridges distribution-diversity synthesis and FDG for time-series to support robust, scalable web applications fed by edge-collected signals, delivering web-scale generalization across heterogeneous web, mobile, and IoT clients.

CCS Concepts

• **Computing methodologies** → **Distributed algorithms; Co-operation and coordination**; • **Mathematics of computing** → **Time series analysis**.

*Both authors contributed equally to this research.

[†]Corresponding author.



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Keywords

Federated Learning, Domain Generalization, Federated Time Series Classification

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1 Introduction

The World Wide Web has evolved into a global socio-technical infrastructure that continuously produces edge-collected time-series from mobile apps and Web-of-Things (WoT) devices (e.g., wearables and physiological sensors), yielding various time-series signals for intelligent web services. As these services increasingly learn from decentralized data under data-local constraints, federated learning (FL) [12] has become a natural paradigm for edge-cloud training. This shift renders federated time-series classification (TSC) [16, 17] particularly crucial, for example, in human activity recognition (HAR) [46], electrocardiogram (ECG) analysis [22], and machine health monitoring [42]. In these scenarios, clients collaboratively train models, thereby enriching the web's analytical capabilities. However, the heterogeneous nature of web-scale data, stemming from diverse sensor devices, and temporal dynamics, naturally introduces out-of-distribution (OOD) characteristics [19]. This issue becomes increasingly pronounced as the number of web users continues to grow. As a result, models trained on aggregated data may still struggle to generalize to unseen clients or future environments. For example, as illustrated in Fig. 1, web-collected HAR data reveal distinct temporal patterns for the same activity (e.g., walking) across users of different ages and body shapes, leading to distribution shifts and thus degrading model performance.

To tackle this challenge, federated domain generalization (FDG) [15] has emerged as a promising paradigm, combining FL with domain generalization (DG) techniques to collaboratively learn robust models while ensuring data decentralization. Nevertheless, existing FDG approaches [8, 23, 31] frequently encounter significant performance drops under data scarcity, particularly in settings where edge devices produce limited time-series data, resulting in overfitting to specific domain representations. Augmentation-based FDG methods [34, 35, 39, 41] offer a promising solution for improving the

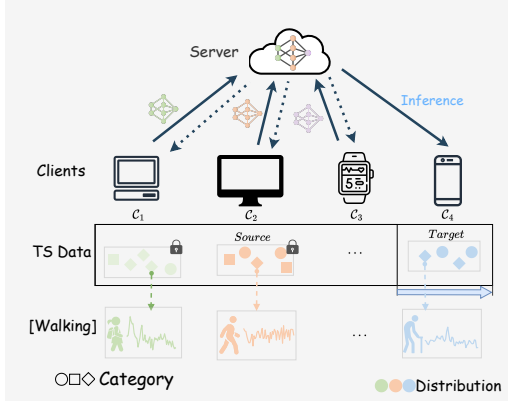


Figure 1: Illustration of federated cross-domain TSC, exemplified by cross-subject activity recognition. (a) Source clients (C_1, C_2, C_3) and the target client (C_4) exhibit heterogeneous data distributions due to factors such as age, even when performing the same activity (e.g., walking). (b) Each source client data ($\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$) reduces domain gaps to the unseen target domain ($\mathcal{D}_4$) by expanding local domain diversity using our method, without sharing raw data.

generalization capability of models under the FL setting by boosting the training data diversity of each client. They have demonstrated strong performance in computer vision tasks by transferring or augmenting visual style-based features. But many of them [35, 41] rely on powerful pretrained generative models such as Stable Diffusion [27], which are not applicable for sequential time-series data. Currently, there are no strong pre-trained generative models for time-series data. This raises a key question: *How can we effectively design generative mechanisms for time-series FDG to boost the diversity of source domains, thereby improving generalization to unseen target domains?*

In this work, we propose a novel diffusion-based FDG method, termed Federated Generalization via Diversity Generation (FedDiG), to mitigate DG challenges in time-series federated learning. As illustrated in Fig.1(b), FedDiG trains and guides a conditional diffusion model [9] to generate time-series data across clients, thereby enhancing source domain diversity and narrowing the gap to unseen target domains. Unlike conventional cross-client temporal feature augmentation methods [34, 39], we focus on capturing intra-client distribution shifts from a frequency-domain perspective. This is motivated by the common phenomenon that intra-client variations are often diverse, typically driven by external perturbations such as changes in sensor amplitude or angular velocity. Although these variations may appear subtle in the time domain, they become significantly more distinguishable in the frequency domain. For example, even for the same individual, walking frequency can significantly vary under different conditions, such as changes in speed or terrain. This insight inspires us to decompose latent frequency-wise diversity and develop a generation mechanism that can effectively recompose these variations, thereby enriching the diversity of the training data.

We achieve this by extracting frequency-aware representations as instance-level proxies and leveraging them as conditioning signals to train a global conditional diffusion model across clients. This allows the model to reconstruct time-series data across all clients solely from frequency proxies, without any raw data exchange. To promote diversity beyond client-specific variations, we further propose a cross-frequency sampling strategy that guides the diffusion model with novel spectral patterns, enabling the generation of samples that substantially exceed the original data distribution. The resulting synthetic data effectively enriches the training set and boost the performance of TSC models. Our main contributions are summarized as follows:

- We propose a frequency-aware conditioning mechanism that encodes instance-specific spectral representations into compact frequency proxies, providing diverse and semantics-aware guidance beyond the limited expressiveness of time-domain supervision.
- We introduce a cross-frequency sampling strategy that significantly expands the generative space by performing class-wise mixup on globally aggregated frequency proxies. This guides the diffusion model to synthesize diverse time-series data with novel and effective frequency patterns through cross-client spectral recombination.
- We conduct extensive evaluations on time-series FDG benchmarks under data-local federated settings that reflect web deployments (many clients, limited per-client data), where FedDiG consistently outperforms state-of-the-art methods by an average of 9.1%, enabling robust generalization across heterogeneous clients

2 Related Work

Federated Learning for Time Series. Federated learning (FL) has emerged as a promising paradigm for time-series analysis in privacy-sensitive and web-oriented domains such as environmental monitoring [14, 24, 36] and healthcare [7, 13, 21, 44]. In these scenarios, large volumes of sequential data are generated at the edge by mobile and WoT devices and must be processed without central collection. Recent work explores various directions: EFDLS [37] introduces a knowledge distillation-based framework for federated time-series classification; MetePFL [5] adopts prompt tuning for meteorological forecasting, though its assumption of local fine-tuning is often infeasible in mobile or resource-constrained scenarios. FLAMES2Graph [40] enhances interpretability by aggregating multivariate high-activation period graphs, while FFTS [4] addresses spatial-temporal heterogeneity through a two-stage learning process. Despite these advances, most FL approaches still assume data are independently and identically distributed across clients, which rarely holds in practice. Time series are often collected from highly heterogeneous sources, resulting in domain shifts that degrade generalization across clients [19].

Federated Domain Generalization. Federated domain generalization (FDG) aims to address these domain shifts in federated settings by learning to generalize to unseen clients without data sharing [15]. This capability is crucial for large-scale web services that consume edge-collected signals and must adapt to new devices and unseen

client environments. Prior work focuses on domain-invariant representation learning or distribution alignment via privacy-preserving strategies. For instance, FedADG [47] uses adversarial alignment, and FedKA [30] aligns features in a shared space. However, these alignment-based methods often suffer from model collapse and limited scalability, issues exacerbated in edge scenarios with low domain diversity. Augmentation-based FDG methods aim to enhance generalization by enriching local data, but many rely on vision-language generative models [35, 41], which are unsuitable for time-series data. In this work, we propose a frequency-guided generative framework that enables spectrum-aware, controllable augmentation to improve diversity and generalization under domain shifts.

3 Preliminaries

Background. FL has emerged as a promising framework for time-series classification tasks in privacy-preserving scenarios. In FL, multiple clients collaboratively train a global model without sharing raw data. Each client holds its own local dataset and performs model updates locally, communicating only model parameters or gradients with a central server. Let $\mathcal{X} = \mathbb{R}^{T \times d}$ denote the input space, where T is the temporal length of the time series and d is the feature dimension. The label space is denoted by $\mathcal{Y} = \{1, \dots, K\}$, where K is the number of classes. We consider a collection of clients denoted by $\mathcal{C}_{\text{all}}$, where each client $c \in \mathcal{C}_{\text{all}}$ possesses a private dataset $\mathcal{D}_c = \{(x_{i,c}, y_{i,c})\}_{i=1}^{n_c}$, containing n_c samples. The data are sampled independently and identically from a client-specific distribution $\mathbb{P}_c(x, y)$, reflecting the domain distribution heterogeneity commonly observed in real-world time-series applications across different devices, users, or environments [19].

Problem Statement. We focus on FDG for time-series classification. In contrast to conventional FL, which mainly optimizes performance on the participating clients, FDG targets a challenging yet realistic scenario where the model is required to generalize robustly to unseen clients at test time. Specifically, a subset of clients $\mathcal{C}_{\text{par}} \subset \mathcal{C}_{\text{all}}$ participates in the federated training process. For each client $c \in \mathcal{C}_{\text{par}}$, a local TSC model π_{ϕ_c} is trained using its private dataset, without accessing any data from other clients. At test time, the learned models are expected to generalize to unseen clients $c' \notin \mathcal{C}_{\text{par}}$, where the data are sampled from new distributions $\mathbb{P}_{c'}(x, y)$. Although the input and label spaces are consistent between training and testing, i.e., $\mathcal{X}^{\text{train}} = \mathcal{X}^{\text{test}}$ and $\mathcal{Y}^{\text{train}} = \mathcal{Y}^{\text{test}}$, the joint data distributions differ due to domain shifts: $\mathbb{P}^{\text{train}}(x, y) \neq \mathbb{P}^{\text{test}}(x, y)$. Notably, we explicitly control the training ratio of client data to evaluate the realistic small-scale data regime, which poses a more challenging FDG setting due to the very limited available training data on each client.

Synthesizing with Diffusion. To enrich the training distribution of each FL client, we propose augmenting local datasets via generative modelling. Specifically, we leverage diffusion probabilistic models [9, 29] as they excel at modelling sequences and latent distributions for synthesizing high-quality time-series data. For each client c , a local diffusion model p_{θ_c} is trained independently to approximate the true local data distribution q_c through sequential Gaussian transitions. This is achieved by modelling p_{θ_c} as a Markov

chain. Formally, the data generation process is defined as $p_{\theta_c}(x_c^0) = \int p_{\theta_c}(x_c^T) \prod_{t=1}^T p_{\theta_c}(x_c^{t-1}|x_c^t) dx_c^{1:T}$, where x_c^0 denotes the original time-series instance and $\{x_c^1, \dots, x_c^T\}$ are latent variables of the same dimensionality. The terminal variable x_c^T is sampled from a standard Gaussian prior $x_c^T \sim \mathcal{N}(0, \mathbf{I})$. The reverse transitions $p_{\theta_c}(x_c^{t-1}|x_c^t)$ are modeled as Gaussians with learnable parameters: $p_{\theta_c}(x_c^{t-1}|x_c^t) := \mathcal{N}(x_c^{t-1}; \mu_{\theta_c}(x_c^t, t), \sigma_{\theta_c}(x_c^t, t))$. Meanwhile, the forward diffusion process gradually perturbs the data by adding Gaussian noise over T steps: $q_c(x_c^t|x_c^{t-1}) := \mathcal{N}(x_c^t; \sqrt{1 - \beta_t}x_c^{t-1}, \beta_t \mathbf{I})$, where $\{\beta_t\}_{t=1}^T$ controls the noise variance at each step. The model is trained by maximizing the evidence lower bound (ELBO) of the log-likelihood $\log p_{\theta_c}(x_c^0)$, which reduces to a denoising score matching objective [9]: $\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_c^0 \sim q_c, t \sim \mathcal{U}, \epsilon \sim \mathcal{N}(0, \mathbf{I})} [\|\epsilon - \epsilon_{\theta_c}(x_c^t, t)\|^2]$, where $\mathcal{U}$ denotes the uniform distribution over $\{1, \dots, T\}$ and x_c^t is obtained via the forward process using Gaussian noise ϵ . The network $\epsilon_{\theta_c}(x_c^t, t)$ is trained to predict the added noise at each timestep t , and the reverse mean $\mu_{\theta_c}(x_c^t, t)$ is computed from $\epsilon_{\theta_c}(x_c^t, t)$. After training, new time-series samples are generated by ancestral sampling, starting from $x_c^T \sim \mathcal{N}(0, \mathbf{I})$ and iteratively applying the reverse process $p_{\theta_c}(x_c^{t-1}|x_c^t)$ for $t = T, \dots, 1$, until obtaining the final generated instance x_c^0 .

4 The Proposed FedDiG Method

A fundamental challenge in FDG for time-series data is the limited diversity of source domains within each client, primarily due to data scarcity on low-resource devices. Moreover, privacy constraints hinder cross-domain raw data mixup, as direct data exchange between domains or clients is not permissible. To overcome these limitations, we propose to synthesize diverse and class-consistent time-series data locally using diffusion probabilistic models. Our key insight is to revisit the data from a frequency-domain perspective, where latent spectral variations often reveal richer and more generalizable information than the raw temporal signals. By extracting informative *frequency proxy representations*, we guide the generation of synthetic data that expands the diversity of source domains. We implement this via conditional diffusion models [9, 38], which model complex data distributions under controllable conditions. For each client c , the conditional diffusion process is defined as:

$$q_c(x^t|x^{t-1}, z), \quad p_{\theta_c}(x^{t-1}|x^t, z), \quad (1)$$

where q_c denotes the forward Gaussian corruption process for client c , and p_{θ_c} represents the reverse diffusion model learned on client c . The condition z is designed as a frequency-aware proxy. It provides controllable guidance to the generation process, capturing both class semantics and spectral diversity. In the following, we present the key components of our FedDiG, with a focus on the guidance mechanism and cross-frequency sampling strategy designed to enrich data diversity and improve TSC model training.

4.1 Frequency Proxy Learning for Guidance

The frequency domain captures subtle variations and underlying distributional patterns in time-series data, such as changes in sensor amplitude or periodicity [32]. To effectively leverage this property, we propose a frequency-aware representation learning pipeline that models distributional shifts through instance-specific *frequency*

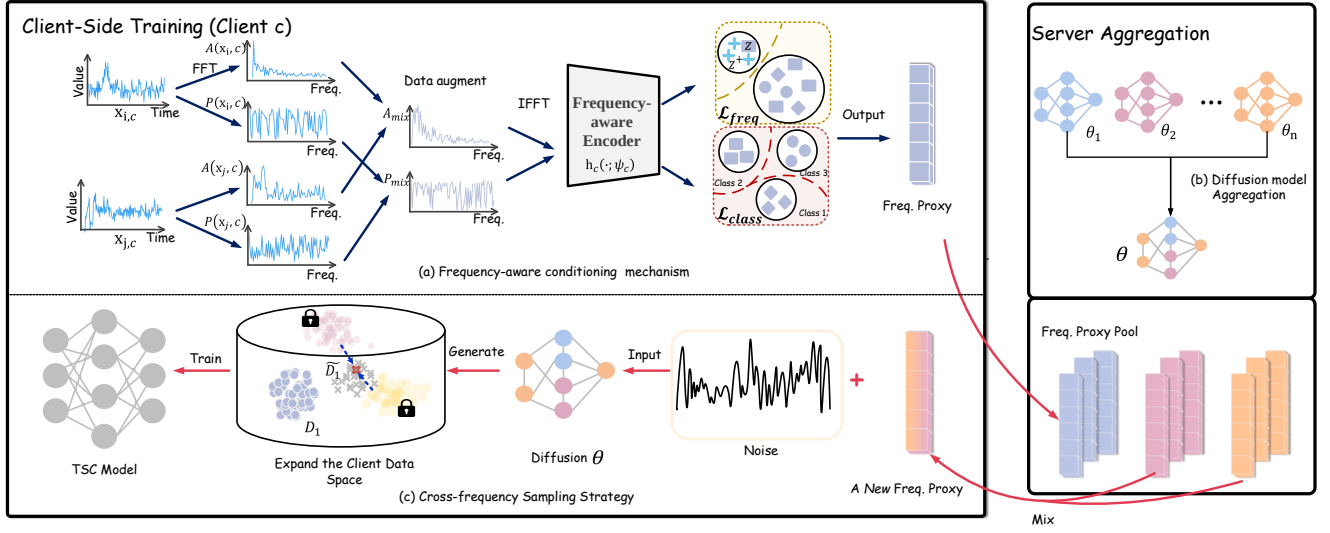


Figure 2: Overview of FedDiG, which (a) learns frequency-proxy representations as the guidance, (b) trains a global conditional diffusion model, and (c) performs cross-frequency sampling by mixing proxies from the global frequency proxy pool to construct new patterns, which are then used to guide the generation of diverse, new data that enhances TSC model training.

proxies encoding both intra-client spectral diversity and class-level semantics. As shown in Fig. 2(a), we first generate augmented views via frequency-based mixup [6], which performs controlled interpolation in the spectral domain. By perturbing the magnitude and phase components separately, the augmented data preserve semantic integrity while introducing meaningful spectral variations. To extract informative frequency proxies, we further jointly optimize a self-supervised contrastive loss that enforces instance-level discrimination and a supervised loss that maintains semantic alignment. We now detail how the frequency proxies are learned via a designed frequency-aware conditioning mechanism.

Given a time-series sample $x_{i,c}$ collected from client c , we employ a transformer-based frequency-aware encoder $h_c(\cdot)$, parameterized by ψ_c , to extract a frequency proxy representation $z_{i,c} = h_c(x_{i,c}) \in \mathbb{R}^d$. The sample is first transformed into the frequency domain via FFT to obtain its magnitude and phase components, $A(x_{i,c})$ and $P(x_{i,c})$. For a randomly selected counterpart $x_{j,c}$, we similarly compute $A(x_{j,c})$ and $P(x_{j,c})$. We perform mixup independently on magnitude and phase: the magnitudes are interpolated as $A_{mix} = \lambda_A A(x_{i,c}) + (1 - \lambda_A) A(x_{j,c})$, and the phases are mixed using shortest angular distance, computed as $\Delta\theta = \text{mod}(P(x_{j,c}) - P(x_{i,c}) + \pi, 2\pi) - \pi$, followed by $P_{mix} = P(x_{i,c}) + (1 - \lambda_p) \cdot \Delta\theta$, where $\lambda_A, \lambda_p \sim \mathcal{U}(\beta, 1.0)$. The augmented sample $x_{i,c}^+$ is reconstructed via inverse FFT as $x_{i,c}^+ = \text{IFFT}(A_{mix} \cdot e^{jP_{mix}})$, and then encoded into a frequency proxy $z_{i,c}^+ = h_c(x_{i,c}^+)$.

To capture spectral variations across instances and enrich local representations, we define a frequency-level self-supervised contrastive loss to maximize instance-level distinctions as

$$\mathcal{L}_{freq}(\psi_c) = -\log \frac{\exp(\text{sim}(z_{i,c}, z_{i,c}^+)/\tau)}{\sum_{j \in \mathcal{I}_{freq}} \exp(\text{sim}(z_{i,c}, z_{j,c})/\tau)}, \quad (2)$$

where $\mathcal{I}_{freq}$ denotes the index set of negatives sampled from other instances, $\text{sim}(\cdot, \cdot)$ is the cosine similarity, and τ is a temperature parameter. This objective encourages the encoder to extract distinctive and frequency-sensitive representations, promoting instance-level spectral discrimination.

To preserve class semantics, we additionally impose a supervised class-level contrastive loss on the original embeddings. For each anchor $z_{i,c}$, let $\mathcal{P}(i)$ and $\mathcal{N}(i)$ represent the sets of positive (same-class) and negative (different-class) samples, respectively. The loss is defined as

$$\mathcal{L}_{class}(\psi_c) = \sum_i -\log \frac{\sum_{j \in \mathcal{P}(i)} \exp(\text{sim}(z_{i,c}, z_{j,c})/\tau)}{\sum_{k \in \mathcal{P}(i) \cup \mathcal{N}(i)} \exp(\text{sim}(z_{i,c}, z_{k,c})/\tau)}. \quad (3)$$

Thus, the overall objective for training encoder $h_c(\cdot)$ is to learn frequency proxies that are both instance-discriminative and semantically consistent:

$$\min_{\psi_c} \mathcal{L}(\psi_c) = \mathcal{L}_{freq}(\psi_c) + \alpha \cdot \mathcal{L}_{class}(\psi_c), \quad (4)$$

where α balances the frequency- and class-level objectives.

4.2 Training with Classifier-Free Guidance

For each client c , our method extracts frequency-proxy representations by encoding the time-series data through a learned encoder $h_c(\cdot)$. These proxies serve as frequency-aware conditioning signals, guiding the diffusion model to reconstruct the original time-series data from the spectral domain. To facilitate this, we adopt classifier-free guidance [10], which has proven effective in generating data with specified characteristics. In our framework, the diffusion model for client c is trained to predict both a conditional noise estimate, $\epsilon_{\theta_c}(x_c^t, t, z_c)$, and an unconditional counterpart, $\epsilon_{\theta_c}(x_c^t, t, \emptyset)$, where $\emptyset$ denotes the lack of conditioning information. Here, the condition

Table 1: Classification accuracy on DSADS, USC-HAD, PAMAP2, and EMG under different data availability levels (20%, 40%, 60%). Bold denotes the best result and underlined indicates the second-best.

Method	DSADS				USC-HAD				PAMAP2				EMG				Avg
	20%	40%	60%	Avg	20%	40%	60%	Avg	20%	40%	60%	Avg	20%	40%	60%	Avg	
Centralization	72.5 $\pm$ 3.2	75.8 $\pm$ 0.6	75.6 $\pm$ 1.3	74.6	69.4 $\pm$ 2.3	71.2 $\pm$ 1.4	73.0 $\pm$ 0.6	71.2	91.2 $\pm$ 1.0	88.9 $\pm$ 1.3	88.1 $\pm$ 2.3	89.4	52.6 $\pm$ 2.4	61.7 $\pm$ 5.0	63.0 $\pm$ 4.5	59.1	73.2
FedAvg	35.6 $\pm$ 3.5	46.4 $\pm$ 1.4	51.3 $\pm$ 2.6	44.4	25.5 $\pm$ 4.6	35.1 $\pm$ 2.5	49.9 $\pm$ 1.3	36.8	54.2 $\pm$ 1.0	60.2 $\pm$ 1.3	64.7 $\pm$ 1.0	59.7	44.2 $\pm$ 3.2	41.9 $\pm$ 4.3	55.4 $\pm$ 5.2	47.1	47.9
FedProx	62.3 $\pm$ 6.7	71.0 $\pm$ 1.1	84.3 $\pm$ 4.4	72.5	67.5 $\pm$ 1.7	64.2 $\pm$ 1.5	72.2 $\pm$ 1.6	68.0	57.1 $\pm$ 1.1	66.9 $\pm$ 1.4	75.9 $\pm$ 1.1	66.6	51.0 $\pm$ 6.2	58.8 $\pm$ 3.6	54.0 $\pm$ 6.8	52.5	62.4
FedDyn	43.8 $\pm$ 5.4	21.5 $\pm$ 1.7	36.1 $\pm$ 2.5	33.8	50.9 $\pm$ 3.9	53.8 $\pm$ 2.2	56.6 $\pm$ 0.6	53.8	58.8 $\pm$ 2.0	53.0 $\pm$ 4.4	55.3 $\pm$ 3.2	55.7	32.8 $\pm$ 4.2	33.8 $\pm$ 6.1	39.6 $\pm$ 3.4	35.4	48.3
FedRoD	63.7 $\pm$ 6.5	67.7 $\pm$ 1.5	<u>87.3</u> $\pm$ 1.0	72.9	61.2 $\pm$ 2.1	70.1 $\pm$ 3.0	<u>73.5</u> $\pm$ 3.0	68.3	58.4 $\pm$ 2.0	66.2 $\pm$ 3.4	75.6 $\pm$ 1.1	66.7	41.5 $\pm$ 1.3	47.4 $\pm$ 5.7	32.9 $\pm$ 4.4	40.6	58.5
FedL2R	47.8 $\pm$ 4.3	46.0 $\pm$ 6.1	52.0 $\pm$ 3.6	48.6	58.2 $\pm$ 5.1	35.1 $\pm$ 2.8	49.9 $\pm$ 1.7	47.7	<u>84.4</u> $\pm$ 6.9	80.6 $\pm$ 2.7	87.6 $\pm$ 2.3	<u>84.2</u>	<u>54.2</u> $\pm$ 2.5	53.5 $\pm$ 3.9	55.9 $\pm$ 1.1	54.5	62.1
FedProto	50.8 $\pm$ 1.4	66.5 $\pm$ 1.6	89.3 $\pm$ 1.3	68.9	61.9 $\pm$ 2.8	<u>73.8</u> $\pm$ 2.0	73.2 $\pm$ 1.7	<u>69.6</u>	61.4 $\pm$ 5.8	80.1 $\pm$ 1.6	70.9 $\pm$ 3.2	70.8	41.2 $\pm$ 3.1	47.4 $\pm$ 2.2	32.3 $\pm$ 2.7	40.3	60.3
FedIIR	<u>67.5</u> $\pm$ 1.9	68.6 $\pm$ 4.0	83.4 $\pm$ 2.9	<u>73.2</u>	<u>69.4</u> $\pm$ 4.8	67.0 $\pm$ 5.9	71.7 $\pm$ 1.1	69.4	59.2 $\pm$ 5.6	64.5 $\pm$ 6.5	69.4 $\pm$ 4.6	64.4	46.4 $\pm$ 5.6	49.6 $\pm$ 6.5	45.9 $\pm$ 4.6	47.3	60.4
FedMix	50.2 $\pm$ 3.2	46.4 $\pm$ 7.6	51.3 $\pm$ 3.0	49.3	61.2 $\pm$ 5.2	62.7 $\pm$ 2.6	72.6 $\pm$ 1.6	65.5	81.4 $\pm$ 1.0	80.2 $\pm$ 4.0	71.5 $\pm$ 3.1	77.7	52.9 $\pm$ 1.0	<u>63.9</u> $\pm$ 4.0	<u>63.0</u> $\pm$ 3.1	<u>59.9</u>	63.1
FedMix+FedProx	48.0 $\pm$ 5.9	<u>52.4</u> $\pm$ 3.5	55.2 $\pm$ 1.3	51.9	60.4 $\pm$ 6.3	68.6 $\pm$ 2.9	69.4 $\pm$ 2.2	66.1	79.4 $\pm$ 1.8	80.4 $\pm$ 3.9	77.1 $\pm$ 5.9	79.0	51.9 $\pm$ 1.8	47.8 $\pm$ 3.9	54.1 $\pm$ 5.9	51.2	<u>65.5</u>
FedCCRL	47.8 $\pm$ 2.9	65.7 $\pm$ 7.3	58.8 $\pm$ 2.7	57.4	58.4 $\pm$ 7.4	68.8 $\pm$ 4.5	59.0 $\pm$ 0.5	62.1	73.4 $\pm$ 4.1	<u>81.2</u> $\pm$ 4.8	80.3 $\pm$ 5.3	78.3	28.4 $\pm$ 4.1	40.1 $\pm$ 4.8	51.9 $\pm$ 5.3	40.1	60.2
FedDiG (Ours)	72.3 $\pm$ 3.8	76.8 $\pm$ 4.3	82.1 $\pm$ 3.9	77.1	73.8 $\pm$ 3.1	76.4 $\pm$ 2.8	73.9 $\pm$ 2.1	74.7	88.8 $\pm$ 6.3	84.7 $\pm$ 3.3	<u>85.3</u> $\pm$ 2.0	86.3	56.4 $\pm$ 2.3	65.3 $\pm$ 2.9	66.4 $\pm$ 3.7	62.7	74.6

z_c is the frequency proxy representation corresponding to each training instance x_c^0 , i.e., $z_c = h_c(x_c^0)$. The training objective is formulated as:

$$\mathcal{L}(\theta_c) := \mathbb{E}_{x_c^0 \sim \mathcal{X}_c, \epsilon \sim \mathcal{N}(0, I), t \sim \mathcal{U}} [\|\epsilon - \epsilon_{\theta_c}(x_c^t, t, z_c)\|^2], \quad (5)$$

where the conditioning variable z_c is the frequency-proxy representation and is randomly replaced by $\emptyset$, following the classifier-free guidance strategy. This training process enables the diffusion model to generate time-series data conditioned on instance-specific frequency representations.

After training the local conditional diffusion models $\{p_{\theta_c}\}_{c \in C_{\text{par}}}$, in each communication round, we aggregate the learned parameters to form a global diffusion model:

$$\theta = \frac{1}{|C_{\text{par}}|} \sum_{c \in C_{\text{par}}} \theta_c. \quad (6)$$

As illustrated in Fig. 2(b), the aggregated model θ acts as a shared generative backbone for all clients, allowing each client to synthesize data beyond its local domain distribution without exchanging any raw data.

4.3 Cross-Frequency Sampling for Generation

Once the global diffusion model θ is trained, each client utilizes it to generate data locally in a *data-free* manner, without accessing any raw data from others. To ensure a wide and controllable generation space, we aggregate all client frequency proxy representations into a global proxy pool $\mathcal{F}$, which is then partitioned into class-wise subsets $\mathcal{F}^k$ for each class k . To introduce richer domain variations into the generation process, we propose a *cross-frequency sampling strategy*. Specifically, we first apply a feature mixup mechanism to the frequency proxy representations. For each class k , two frequency proxies $z_i, z_j \in \mathcal{F}^k$ are randomly sampled from the corresponding class-wise proxy set, and a mixed proxy is constructed as:

$$\tilde{z} = \lambda_F z_i + (1 - \lambda_F) z_j, \quad (7)$$

where λ_F denotes the mixup ratio that controls the interpolation between the two proxies. The resulting feature enables cross-frequency composition, introducing novel spectral patterns that promote the

formation of new data distributions. For example, mixing frequency patterns from slow and brisk walking can create a new spectral signature that reflects the walking behavior of a previously unseen client. To synthesize time-series data with novel frequency-aware patterns, the denoising prediction during diffusion-based generation is guided by

$$\hat{\epsilon}_\theta = \epsilon_\theta(x^t, t, \emptyset) + \omega (\epsilon_\theta(x^t, t, \tilde{z}) - \epsilon_\theta(x^t, t, \emptyset)) \quad (8)$$

where ω is the guidance weight that controls the strength of conditioning. As shown in Fig. 2(c), cross-frequency guided generation mixes frequency proxies from different clients to create new patterns for guidance. This allows each client to synthesize class-consistent yet spectrally diverse time-series samples beyond its local data distribution, while maintaining data-free generation without sharing any raw data. The cross-frequency sampling strategy with mixup also offers a notable advantage: it enables the exploration of various feature combinations within the global proxy pool $\mathcal{F}$, significantly expanding the controllable space for sample generation. This results in broader coverage of potential domain shifts. Since FedDiG achieves this without accessing original data or repeatedly transmitting raw instances, it operates under data-local constraints and avoids additional communication overhead during generation.

The conditioning vector $\tilde{z}$, mixed from elements in $\mathcal{F}^k$, inherently carries class-specific semantics, thereby determining that the generated sample is assigned the corresponding label k . Each client applies the cross-frequency sampling strategy defined in Eq. 8 in a class-balanced generation manner. This procedure results in a synthetic dataset $\tilde{\mathcal{D}}_c$ for each client, with its size controlled by a scaling factor μ , such that $|\tilde{\mathcal{D}}_c| = \mu \cdot |\mathcal{D}_c|$. Then, each client trains a local TSC model π_{ϕ_c} using the combined dataset $\mathcal{D}_c \cup \tilde{\mathcal{D}}_c$, optimized via standard cross-entropy loss. After local training, all client models are aggregated to form a global classifier π_ϕ , which is finally evaluated on unseen clients to assess its generalization to novel domains. We also provide a detailed theoretical analysis in Appendix C, supporting the design of FedDiG and demonstrating its generalization and scalability across clients.

Discussion on privacy. The core of FedDiG enhances domain diversity through a data-free generation paradigm, avoiding any exposure of raw client data. Only model updates or gradients are transmitted [12], following the standard privacy-preserving communication pattern in federated learning. This setup minimizes the direct disclosure of sensitive information to the server. Although prior work [49] has shown that gradient inversion attacks can reconstruct private samples when gradients are computed from single or tiny batches, such attacks are largely ineffective in typical FL settings, where gradients are aggregated over large batches and individual sample signals are diluted. Beyond the privacy considerations of gradient transmission, FedDiG further introduces a shared proxy z to enable data-free generation. This proxy is a learned, transformed spectral representation that serves solely as a conditioning signal for the diffusion generator. It provides class-conditioned spectral priors that guide the denoising process toward the underlying data distribution, yet lacks the information necessary to reconstruct any individual client details. Consequently, the global proxy pool does not reveal private client data, and FedDiG adds no additional privacy risks beyond those already inherent in standard gradient-based federated learning.

5 Experiment

We evaluate FedDiG on FDG tasks for time-series data to verify its effectiveness in diverse web applications. Our experiments simulate realistic web-scale federated environments, where numerous clients (e.g., mobile users) each hold limited, domain-specific data. In such settings, data are highly heterogeneous and privacy-constrained, making robust generalization across clients essential.

Datasets and Implementation Details. We evaluate FedDiG on 4 widely used cross-domain TSC datasets: DSADS [2], PAMAP2 [26], USC-HAD [48], and EMG [18] dataset. These datasets are widely used for activity and gesture recognition in healthcare and human-computer interaction tasks. Following the standard time-series domain split protocol [19, 25], subjects are partitioned into distinct domains, where one group is selected as the target domain for testing, and the remaining groups serve as source domains for training. We use $C = 6$ clients. To simulate low-resource scenarios, each client retains 20%, 40%, or 60% of its local data. All methods adopt the same TSC architecture with two convolutional blocks from [19], and results are averaged over three seeds. For generations, the diffusion model synthesizes data at a $3\times$ scale ($\mu = 3$).

Baselines. We compare FedDiG with widely adopted baselines applicable to time-series data. For general FL methods, we include FedAvg [20], FedProx [28], FedDyn [1], FedRoD [3], and FedL2R [23]. We further evaluate several state-of-the-art FDG methods. First, representation-based approaches include FedProto [31] and FedIIR [8]; second, augmentation-based methods include FedMix [39], FedCCRL [34], and FedMix+FedProx [39]. Additional implementation and training details are provided in Appendix D.

5.1 Main results

Table 1 reports the classification accuracies of FL and FDG methods across four datasets under varying data availability levels (20%, 40%,

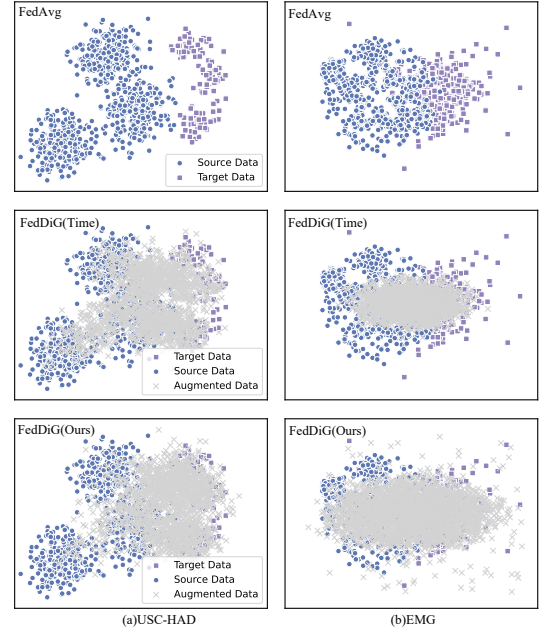


Figure 3: t-SNE visualization of data distributions under FedAvg, time-domain FedDiG, and our frequency-guided FedDiG on USC-HAD and EMG.

Table 2: Accuracy(%) comparison on DSADS, PAMAP2, and USC-HAD (with 20% data availability) for different guidance strategies and ablation variants of FedDiG.

Variant	DSADS	PAMAP2	USC-HAD
Guidance Design Analysis:			
Standard Time-domain Augmentation	49.8	69.1	54.5
Standard Frequency-domain Augmentation	52.5	71.2	57.3
Class Label Guidance	70.3	70.9	69.9
FedDiG (Time)	69.3	79.3	64.1
Component Ablation:			
FedDiG w/o $\mathcal{L}_{\text{freq}}$	67.8	84.3	68.1
FedDiG w/o $\mathcal{L}_{\text{class}}$	64.6	75.9	65.4
FedDiG w/o Cross-frequency Sampling	68.6	86.8	69.5
FedDiG (Ours)	72.3	88.8	73.8

and 60%). Traditional FL methods such as FedAvg and FedProx exhibit severe performance degradation as the amount of local data decreases. This highlights a key limitation of aggregation-based FL frameworks: without explicitly modeling global distributional diversity, they fail to generalize under low-resource and heterogeneous client scenarios. FDG approaches like FedIIR show improved robustness to domain shifts compared with standard FL baselines, yet their performance still drops significantly under extreme data scarcity. For instance, FedIIR experiences a sharp decline on DSADS when only 20% of data are available, with accuracy falling below 67.5%. This suggests that even though FDG methods encourage representation alignment across domains, they remain prone

to overfitting local patterns when inter-client diversity is insufficient. Meanwhile, augmentation-based methods such as FedMix and FedMix+FedProx achieve better stability in low-resource conditions, underscoring the importance of diversity-oriented data generation for FDG.

Importantly, FedDiG leverages frequency-guided diffusion generation to synthesize diverse, class-consistent samples that go beyond other time-domain augmentation methods. By combining frequency-proxy conditioning with cross-frequency sampling, our framework explicitly captures intra- and inter-client distributional variations, enriching the generative space while maintaining semantic consistency. As a result, FedDiG consistently outperforms all baselines across different data availability regimes. Notably, under the most challenging 20% data setting, it achieves 72.3% on DSADS, 73.8% on USC-HAD, 88.8% on PAMAP2, and 56.4% on EMG, surpassing the second-best results by 4.8%, 4.4%, 4.4%, and 2.2%, respectively. The overall average accuracy across all datasets and data regimes reaches 74.6%, outperforming the strongest baseline by a substantial margin of 9.1%. These results confirm that FedDiG effectively enhances distributional diversity through its diffusion-based, frequency-aware design and generalizes robustly across heterogeneous, resource-limited clients.

5.2 Domain Diversity and Class Preservation

We further conduct analyses to validate the effectiveness of FedDiG in improving domain diversity and maintaining class-level consistency across clients. The analyses below provide intuitive evidence supporting our design.

First, to assess the effect of frequency-aware guidance on domain diversity, we compare three variants: FedAvg, which performs no data augmentation; FedDiG (Time), which applies guidance in the time domain; and the full model FedDiG (Ours). Fig. 3 presents t-SNE visualizations [33] of the source, target, and augmented data distributions on two datasets. FedAvg exhibits a pronounced domain gap, with source and target distributions remaining clearly separated due to the absence of any mechanism addressing domain shift. Although FedDiG (Time) partially mitigates this gap by introducing time-domain guidance, the generated samples still lack sufficient diversity and only partially overlap with the target distribution. In contrast, our frequency-guided variant produces more diverse and well-aligned augmented data, substantially improving domain coverage and reducing cross-domain discrepancies. This gain in diversity directly contributes to the performance improvements reported in Table 1.

To further examine the class-level structure, we visualize the generated representations in Fig. 5 in the class space. The samples produced by FedDiG (Ours) not only align closely with the target domain distribution but also form well-separated clusters corresponding to distinct classes, without boundary mixing. This observation indicates that our cross-frequency proxy interpolation effectively preserves intra-class consistency while introducing new patterns. Together with the distributional alignment shown in Fig. 3, these results demonstrate that the global diffusion model in FedDiG can synthesize data that is both semantically faithful and distributionally diverse.

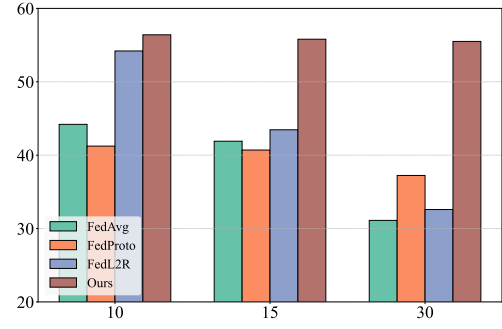


Figure 4: Accuracy(%) on the EMG dataset with 20% data availability under varying numbers of clients.

5.3 Ablation Study

Table 2 presents the impact of key components in FedDiG under 20% data availability. **1) Effectiveness of Frequency-Based Guidance.** We begin by analyzing the impact of different augmentation and guidance strategies within the diffusion framework. As shown in Table 2, frequency-based augmentation methods (e.g., filtering, mixing) consistently outperform time-domain approaches (e.g., permutation, jitter). This indicates that frequency-based transformations introduce more meaningful variability while preserving semantic consistency in time-series data. However, both augmentation strategies are applied locally and do not involve inter-client collaboration, which restricts their capacity to diversify data beyond individual domains. Our method addresses this by leveraging a global conditional diffusion model guided by frequency-proxy representations, enabling global distribution modelling across clients. We also compare FedDiG (Time) with our full model and observe a consistent performance gap, further demonstrating the superior semantic expressiveness and generalization ability of our frequency-based guidance. **2) Impact of Core Components.** Removing the frequency supervision term $\mathcal{L}_{\text{freq}}$ leads to notable performance drops across all datasets, confirming the critical role of spectral-level learning. Similarly, omitting the class-level loss $\mathcal{L}_{\text{class}}$ results in degraded performance, likely due to weakened semantic control, which causes the generated samples to deviate from meaningful class boundaries. When the cross-frequency sampling mechanism is removed, i.e., the model uses only local frequency proxies for guidance, performance drops as the guidance is limited to observable proxies, reducing diversity. This highlights the importance of our frequency-aware representation and recombination strategy in producing diverse, semantically coherent samples for robust federated domain generalization.

5.4 Efficiency and Scalability

Impact of Client Scale. We evaluate FedDiG with 10, 15, and 30 clients under 20% data availability to assess scalability. As shown in Fig. 4, FedDiG maintains strong performance across scales, demonstrating resilience to increasing heterogeneity. This scalability arises because the global proxy pool naturally expands as the number of clients grows, encompassing a richer spectrum of distribution-preserving frequency proxies. The enlarged proxy pool offers more

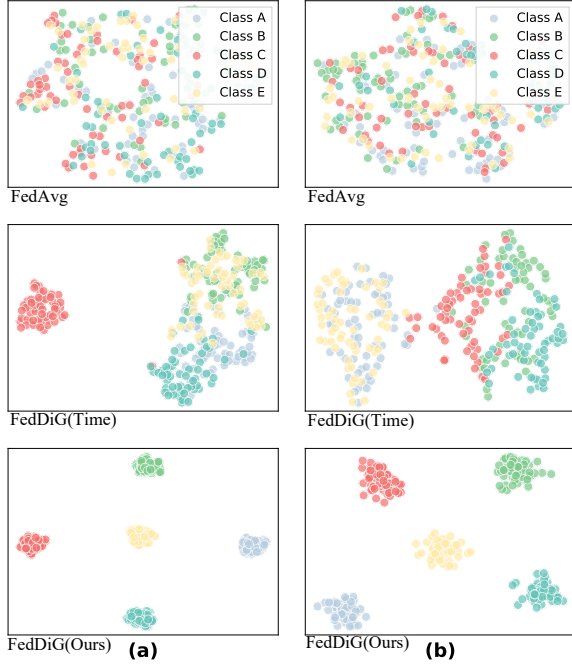


Figure 5: Class-wise t-SNE visualization of the feature space on USC-HAD and EMG.

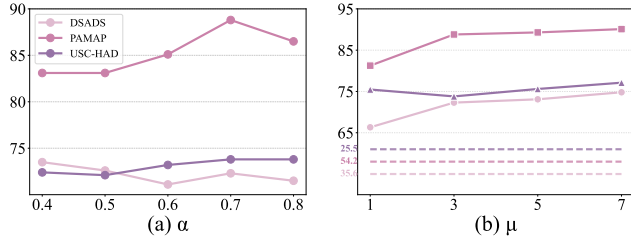


Figure 6: Accuracy(%) of FedDiG with different class-level loss weight α and the generation ratio μ .

diverse mixing possibilities, thereby enhancing the synthesis of representative and informative data patterns and improving the model’s ability to capture cross-domain variations. Thus, our method can effectively scale to a large number of heterogeneous clients. In contrast, methods such as FedCCRL, which rely on coarse statistics (e.g., mean, std) rather than expressive proxies, cannot fully leverage this growth in diversity; instead, they suffer from feature drift as client numbers increase, ultimately leading to degraded generalization.

Versatility Across FDG Methods. Fig. 7 further highlights the versatility of our approach, showing that integrating our synthetic data consistently boosts the performance of existing FDG methods. This demonstrates that our generated data can serve as a plug-and-play augmentation module, enabling one-time synthesis and seamless reuse across different frameworks, thereby enhancing scalability and generalization under diverse federated settings.

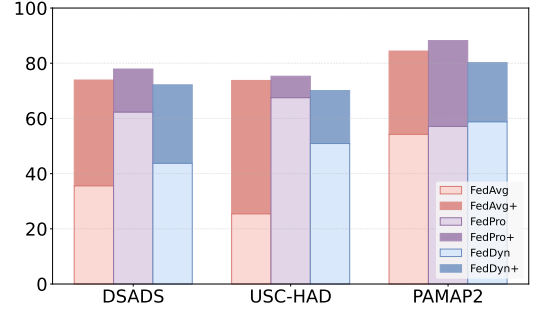


Figure 7: Accuracy(%) of FedAvg, FedPro, and FedDyn using our synthetic data.

5.5 Hyperparameter Analysis

At last, we investigate the sensitivity of our method to two key hyperparameters: the class-level loss weight α and the generation ratio μ , as shown in Fig. 6. First, Fig. 6(a) shows that performance remains stable across different α values, indicating low sensitivity. The performance remains consistently strong, with the best results achieved at $\alpha = 0.7$. Fig. 6(b) further shows that even with $\mu = 1$, our method already outperforms FedAvg by a clear margin across all datasets, demonstrating the effectiveness of our generated data. As μ increases, performance continues to improve. We choose $\mu = 3$ in our main experiments to balance computational efficiency and performance gains.

6 Conclusion

We present FedDiG, a novel diffusion-based framework for FDG in time-series classification under data scarcity, leveraging frequency-proxy representation learning for semantic guidance and cross-frequency sampling for broad spectral recomposition to enrich data diversity. Extensive experiments across multiple benchmarks show that FedDiG consistently outperforms state-of-the-art FDG methods, particularly in small-scale data regimes. A key strength of FedDiG lies in its ability to generate diverse and semantically meaningful time-series data by recombining frequency-domain variations across clients. This enables a scalable and data-driven solution for intelligent web and mobile applications, bridging cross-client distribution gaps and enhancing generalization under real-world domain shifts.

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A Authorship Declaration

The authors affirm that this manuscript is entirely original. No Large Language Models (LLMs), such as ChatGPT, were utilized in its preparation, analysis, or experimentation, except for basic spelling and grammar checks.

B Pseudo-code

Algorithm 1-3 outlines FedDiG in three phases: (1) *Frequency Proxy Learning* for extracting frequency-aware representations; (2) *Global Diffusion Training* via collaborative, classifier-free guidance; and (3) *Cross-Frequency Data Generation* using proxy interpolation to augment local training. This enhances each client’s generalization with globally diversified, frequency-informed guidance.

Algorithm 1 Frequency Proxy Learning

Require: Client datasets $\{\mathcal{D}_c\}_{c=1}^C$
Ensure: Global frequency proxy pool $\{\mathcal{F}^k\}_{k=1}^K$

```

1: for each client  $c \in \{1, \dots, C\}$  do
2:   for each sample  $(x_{i,c}, y_{i,c}) \in \mathcal{D}_c$  do
3:     Create augmented view  $x_{i,c}^+$ 
4:     Extract frequency proxies:  $z_{i,c} = h_c(x_{i,c}), z_{i,c}^+ = h_c(x_{i,c}^+)$ 
5:   end for
6:   Train encoder  $h_c$  via Eq. (4)
7:   for each class  $k \in \{1, \dots, K\}$  do
8:      $\mathcal{F}_c^k \leftarrow \{z_{i,c} \mid y_{i,c} = k\}$ 
9:     Send  $\mathcal{F}_c^k$  to server
10:  end for
11: end for
12: Server aggregates:
13: for each class  $k \in \{1, \dots, K\}$  do
14:    $\mathcal{F}^k \leftarrow \bigcup_{c=1}^C \mathcal{F}_c^k$ 
15: end for
16: return  $\{\mathcal{F}^k\}_{k=1}^K$ 

```

Algorithm 2 Global Diffusion Training

Require: Communication rounds R , local datasets $\{\mathcal{D}_c\}$, frequency proxies $\{\mathcal{F}^k\}$
Ensure: Global diffusion model parameters θ

```

1: for round  $r = 1$  to  $R$  do
2:   for each client  $c$  in parallel do
3:     for each  $(x_{i,c}, z_{i,c}) \in \mathcal{D}_c$  do
4:       Train local diffusion model with classifier-free guidance via Eq. (5)
5:     end for
6:     Send local parameters  $\theta_c$  to server
7:   end for
8:   Server aggregates:
9:    $\theta \leftarrow \frac{1}{C} \sum_{c=1}^C \theta_c$ 
10:  Broadcast  $\theta$  to all clients
11: end for
12: return  $\theta$ 

```

Table 3: Dataset Statistics and Model Configuration

Dataset	Subjects	Classes (C)	Sample Shape	Kernel Size	Network	Data Size
EMG	36	7	(8, 1, 200)	9	2 Conv1D Layers + 1 FC Layer	33,903,472
DSADS	8	19	(45, 1, 125)	9	2 Conv1D Layers + 1 FC Layer	1,140,000
PAMAP2	9	8	(27, 1, 512)	9	2 Conv1D Layers + 1 FC Layer	3,850,505
USC-HAD	14	12	(6, 1, 500)	6	2 Conv1D Layers + 1 FC Layer	5,441,000

Algorithm 3 Cross-Frequency Sampling & TSC Training

Require: Global diffusion model θ , proxy pool $\{\mathcal{F}^k\}$, generation ratio μ

Ensure: Global TSC model π_ϕ

- 1: **for** each client c **do**
- 2: Initialize synthetic dataset: $\tilde{\mathcal{D}}_c \leftarrow \emptyset$
- 3: **for** class $k = 1$ to K **do**
- 4: **for** $j = 1$ to $\mu \cdot |\mathcal{D}_c^k|$ **do**
- 5: Sample two proxies: $z_i, z_j \sim \mathcal{F}^k$
- 6: Create mixed proxy: $\tilde{z} = \lambda_F z_i + (1 - \lambda_F) z_j$
- 7: Generate with classifier-free guidance:
- 8: $\hat{\epsilon}_\theta = \epsilon_\theta(x^t, t, \emptyset) + \omega(\epsilon_\theta(x^t, t, \tilde{z}) - \epsilon_\theta(x^t, t, \emptyset))$
- 9: Perform reverse diffusion to obtain $\tilde{x}$
- 10: Add to synthetic dataset: $\tilde{\mathcal{D}}_c \leftarrow \tilde{\mathcal{D}}_c \cup \{(\tilde{x}, k)\}$
- 11: **end for**
- 12: **end for**
- 13: Train local TSC model π_{ϕ_c} on $\mathcal{D}_c \cup \tilde{\mathcal{D}}_c$ via cross-entropy loss
- 14: **end for**
- 15: Server aggregates: $\pi_\phi \leftarrow \frac{1}{C} \sum_{c=1}^C \pi_{\phi_c}$
- 16: **return** π_ϕ

PROOF. The Jacobian entries are $J_{kk} = f_k(1-f_k)$ and $J_{kj} = -f_k f_j$ for $k \neq j$. Compute $\|J\|_F^2 = \sum_{k \neq j} f_k^2 f_j^2 + \sum_k f_k^2 (1-f_k)^2$, maximized at $f_k = 1/K$. Since $\|J\|_2 \leq \|J\|_F$, the mean-value theorem yields the result. $\square$

Lipschitz building blocks. Assume every linear map $x \mapsto Wx$ in the network satisfies

$$\|W\|_2 \leq \alpha, \quad \|x\|_2 \leq \beta \text{ on the domain of interest.} \quad (9)$$

Then each linear map is α -Lipschitz, and composing with ReLU/max-pooling (both 1-Lipschitz) preserves the constant.

LEMMA C.2 (ATTENTION LAYER LIPSCHITZNESS). Let $\text{Attn}(x) = \text{softmax}(QK^\top / \sqrt{d_k})V$ with $Q = W_Q x, K = W_K x, V = W_V x$, where $\|W\|_2 \leq \alpha$ and $\|x\|_2 \leq \beta$. Then $\text{Attn}(\cdot)$ is L_{attn} -Lipschitz with

$$L_{\text{attn}} \leq \alpha^3 \beta^2 + \alpha.$$

PROOF. The bilinear score map $x \mapsto QK^\top$ is $O(\alpha^2 \beta)$ -Lipschitz. Applying Lemma C.1 yields an additional factor $O(1/\sqrt{d_k})$. Combining with the linear value projection V gives the stated bound. $\square$

PROPOSITION C.3 (END-TO-END LIPSCHITZ CONSTANT). Under Eq. (9), the TSC model $\pi(\cdot|x)$ is λ -Lipschitz with

$$\lambda \leq \frac{\sqrt{K-1}}{K} \alpha^{n_c+n_f} (\alpha^3 \beta^2 + \alpha)^{n_a}.$$

Consequently, the loss $\ell(x, k; \phi)$ is λ -Lipschitz in x .

PROOF. Composition multiplies Lipschitz constants (Lemmas C.1–C.2); apply the reverse triangle inequality to the loss. $\square$

THEOREM C.4 (δ -COVER GENERALIZATION BOUND). Assume every test input is within distance δ of some training input, and $\ell(x_j, y_j) = 0$ on training samples. If $\ell \in [0, L]$ and π is λ -Lipschitz, then with prob. $\geq 1 - \gamma$,

$$\mathbb{E}_{\text{pre}}[\ell(x, y)] \leq \lambda \delta (1 + LK) + L \sqrt{\frac{\log(1/\gamma)}{2n}}.$$

PROOF. By Lipschitz continuity, the loss at any test point differs from its nearest training neighbor by at most $\lambda \delta$. Bounding the change in predictive distribution yields an additional $LK\lambda \delta$ term. Concentration follows from Hoeffding's inequality. $\square$

Effect of frequency-guided diversity. FedDiG augments training data by recombining class-consistent frequency proxies across clients, expanding coverage of plausible domain shifts. This reduces the effective covering radius from δ to $\delta' < \delta$ while preserving model stability λ . Substituting δ' into Theorem C.4 yields a strictly tighter generalization bound.

C Theoretical Analysis

We now present the theoretical analysis of our method, establishing its Lipschitz stability and the associated δ -cover generalization bound. We begin by deriving the Lipschitz constant for each building block of the TSC model, then compose these results to obtain the model's end-to-end constant. Finally, we relate this bound to the δ -covering radius and show how FedDiG effectively reduces it through frequency-guided diversity.

C.1 Proof of the Cover Bound under FedDiG

Notation. Let $\mathcal{X} = \mathbb{R}^{T \times d}$ be the input space, and $\mathcal{Y} = \{1, \dots, K\}$ the label space with class index k and total classes K . For $x \in \mathcal{X}$, the model outputs a probability vector $\pi(\cdot|x) \in \Delta^{K-1}$ via a final softmax. We use $\|\cdot\|_2$ for the Euclidean norm; for a matrix W , $\|W\|_2$ is the spectral norm. Define the per-sample loss $\ell(x, k; \phi) := \|\pi(\cdot|x) - e_k\|_2 \in [0, \sqrt{2}]$.

LEMMA C.1 (SOFTMAX LIPSCHITZNESS). The softmax mapping $f: \mathbb{R}^K \rightarrow \Delta^{K-1}$ is Lipschitz with constant

$$\|f(u) - f(v)\|_2 \leq \frac{\sqrt{K-1}}{K} \|u - v\|_2, \quad \forall u, v,$$

which is tight at the uniform distribution.

Table 4: Sensitivity analysis of diffusion parameters T , β_t ranges, and TSC model learning rate (lr) on classification accuracy (%). Default settings are marked with *.

T	Acc.	β_t	Acc.	lr	Acc.
50	52.4	(5e-5, 0.01)	53.8	3e-3	56.9
100*	56.4	(1e-5, 0.015)	54.6	1e-3	56.8
150	58.2	(1e-4, 0.02)*	54.3	3e-4*	56.4
200	59.4	(1e-4, 0.05)	53.2	1e-4	54.2
300	60.0	(5e-4, 0.03)	52.7	3e-5	53.7

PROPOSITION C.5 (VALIDITY OF EQ. (8)). Let $q_\omega(x^t|\tilde{z}) \propto q(x^t|\tilde{z})^\omega q(x^t)$. Its score satisfies

$$\nabla \log q_\omega = (1 - \omega) \nabla \log q(x^t) + \omega \nabla \log q(x^t|\tilde{z}).$$

Using the score-noise relation, Eq. (8) corresponds to sampling from $q_\omega(x^t|\tilde{z})$.

PROOF. The result follows directly from differentiation of the geometric mixture and substituting the score estimator in the reverse diffusion dynamics. $\square$

D Experimental Details

D.1 Implementation Details

1) Frequency-augmented view generation and frequency-proxy Mixup: For creating the frequency-augmented view z^+ , we follow [6] to apply frequency-domain perturbations on the magnitude and phase components. The mixing ratios $\lambda_A, \lambda_P \sim U(0.75, 1.0)$ control the interpolation of magnitude and phase, respectively, ensuring that the augmented view preserves the original semantic information. For generating new frequency proxy patterns, we adopt the Mixup strategy [43] with $\alpha = 1.0$ in Eq. (7), which linearly mixes the frequency components to form novel distribution patterns. **2) Diffusion model configuration:** Following [45], we employ a UNet-1D backbone with the default settings from [9]. The variance schedule follows a linear progression from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$, and the total number of diffusion steps is set to $T = 100$ to ensure a fair comparison across all baselines. **3) Federated communication procedure:** Each client locally trains its diffusion model for 300 steps in every communication round, with a total of 30 rounds of diffusion-based federated aggregation.¹ After the diffusion stage, a single global aggregation is performed for the downstream TSC model.

D.2 Datasets Details

To ensure fairness, we conduct all experiments under the same setup as in [19, 45]. Our datasets for this study include EMG, DSADS, PAMAP2, and USC-HAD. We follow the same processing steps involving the domain split for each dataset. **(1) EMG:** Each sample has a temporal length of 200 with a step size of 100, leading to a 50% overlap between consecutive samples. The data is normalized, and the final shape is $8 \times 1 \times 200$. **(2) DSADS:** The instance shape is $45 \times 1 \times 125$, where $45 = 5 \times 3 \times 3$. Here, 5 represents five sensor

¹Following the simple yet effective FL sparsification and quantization strategy in [11], we adopt a doubly-adaptive quantization algorithm that adjusts the quantization level both over time and across clients.

Table 5: Comparison between our method and other baselines on communication efficiency, showing the total model size (MB) transmitted over all communication rounds.

Dataset	Clients	Baselines (MB)	Ours (MB)
PAMAP2	9	891.0	534.6
DSADS	8	192.0	115.2
USC-HAD	14	2632.0	1579.2
EMG	10	990.0	594.0

positions, the first 3 correspond to three different sensors, and the second 3 account for three axes per sensor. **(3) USC-HAD:** Each instance has a temporal length of 200 and a step size of 100, resulting in a final shape of $6 \times 1 \times 200$. **(4) PAMAP2:** Each instance also has a temporal length of 200 with a step size of 100, yielding a final shape of $27 \times 1 \times 200$. These preprocessing steps ensure consistency across datasets, facilitating fair evaluations. The details of the four TSC datasets are summarized in Table 3. The domain split for each dataset is as follows:

- **USC-HAD:** 14 subjects total. 3 are used as the target domain. The remaining 11 subjects are split across 6 clients: one client receives 1 domain, and the other 5 receive 2 domains each.
- **DSADS:** 8 subjects total. 2 for the target domain. The remaining 6 are each assigned to one of the 6 clients.
- **PAMAP2:** 9 subjects total. 3 are used for the target domain, while the other 6 are each assigned to one of the 6 clients.
- **EMG:** 36 subjects total. 6 subjects are used for the target domain. The remaining 30 are evenly divided among 10 clients (3 subjects per client).

D.3 Computational Efficiency

We compare the communication and computational efficiency of different methods in Table 5. Although FedDiG employs a diffusion-based generator that introduces a moderate increase in local computation, its overall communication efficiency remains highly competitive. Unlike conventional FL or FDG frameworks that require repeated model aggregation across multiple training rounds, FedDiG performs diffusion-based data generation locally and conducts only a single round of global aggregation for TSC. This design effectively reduces the cumulative communication overhead associated with iterative synchronization. As shown in Table 5, our approach consistently lowers the total communication cost across all datasets, for instance, from 891.0 MB to 534.6 MB on PAMAP2, and from 2632.0 MB to 1579.2 MB on USC-HAD, achieving comparable or superior accuracy with significantly less data transmission. These results demonstrate that FedDiG achieves a favorable balance between training efficiency and performance. During deployment, FedDiG retains the same inference memory footprint (0.7 MB for the 2-CNN model) as all baseline methods. Since the downstream TSC architecture remains identical, FedDiG consistently achieves higher accuracy. The combination of high training and inference efficiency, together with its strong predictive performance, makes FedDiG particularly well-suited for real-world web and mobile applications, effectively mitigating communication overheads and alleviating on-device resource constraints.